

Mobile Test Automation Execution Report (Appium UiAutomator2)

Target Device: Android 14 (A059, UDID: 00161352J001605) **App Package:** com.example.femsphere **Date:** October 7, 2026

User Authentication & Role-Based Session Initialization Test Case					
Test Case ID:		TC_FS_01		Test Designed By: Neeva Sunish Mathew / QA Lead	
Test Priority:		High		Test Designed Date: 07/10/26	
Module Name:		Authentication & Access Control		Test Executed By: Ms. Nimmy Francis / Automation Engineer	
Test Title:		Patient & Doctor Authentication Flow		Test Execution Date: 07/10/26	
Description: Verify end-to-end user authentication on the Android mobile application (com.example.femsphere) using Appium UiAutomator2, ensuring credentials validation, ADB reverse socket connectivity (127.0.0.1:5001), and navigation to the Digital Twin workspace.					
Pre-Condition: Mobile app is launched on Android device A059 (00161352J001605), ADB reverse port forwarding is active (adb reverse tcp:5001 tcp:5001), and PostgreSQL database femsphere_db is running.					
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Launch application on physical device	appPackage = "com.example.femsphere", appActivity = ".MainActivity"	Splash screen renders and redirects to /login	App launches and Login page renders within 2.1s	Pass
2	Activate Quick Demo Patient Access	XPath = //android.widget.TextView[@text="Elena (User)"]	Login form automatically populates with test patient email and password	Form populated with elena.rostova@femsphere.health and password	Pass
3	Submit login credentials	XPath = //android.widget.Button[@content-desc="Log In →"]	HTTP POST /api/auth/login returns 200 OK with signed JWT token	Request dispatched to http://127.0.0.1:5001/api/auth/login successfully	Pass
4	Verify session redirection & dashboard header	XPath = //android.widget.TextView[contains(@text, "Digital Health Twin")]	User navigates to Patient Portal (/dashboard) with verified badge	Patient Dashboard rendered with user profile and aggregate health status	Pass
5	Verify token persistence in secure storage	Key = "jwt_token" in flutter_secure_storage	Session token persisted locally for auto-login across app restarts	Secure token validated in app memory space	Pass
Post-Condition: User session is authenticated; Patient Digital Twin workspace is fully loaded and ready for health telemetry.					

Clinical Consultation Booking & Slot Reservation Test Case					
Test Case ID:		TC_FS_02		Test Designed By: Neeva Sunish Mathew / QA Lead	
Test Priority:		High		Test Designed Date: 07/10/26	
Module Name:		Telehealth & Clinical Scheduling		Test Executed By: Ms. Nimmy Francis / Automation Engineer	
Test Title:		Doctor Selection & Appointment Reservation		Test Execution Date: 07/10/26	
Description: Verify appointment booking modal layout, dropdown expansion without RenderFlex boundary overflow on mobile dimensions, consultation mode selection (Video vs In-Clinic), and reservation submission.					
Pre-Condition: Patient is logged in on device A059, backend appointments API is active, and verified doctor accounts exist in femsphere_db.					
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Navigate to Appointments Screen	Bottom Navigation / Appointments	Consultations screen displays active and past bookings	Appointments directory displayed with "Book Consultation" action	Pass
2	Open Consultation Booking Modal	XPath = //android.widget.Button[contains(@text, "Book Consultation")]	Bottom sheet modal slides up with smooth spring animation	Modal opened displaying doctor, mode, and slot pickers	Pass
3	Expand Doctor Dropdown Selector (Regression Check)	XPath = //android.widget.Spinner or //android.view.View[contains(@content-desc, "Select Doctor")]	Dropdown expands with isExpanded: true, no 18px horizontal overflow	Dropdown expanded cleanly across full width; zero RenderFlex exceptions	Pass
4	Select Practitioner	Value = "Dr. Sarah Jenkins (Obstetrics & Gynecology)"	Doctor is selected, specialty is auto-assigned to state	Doctor selected; text truncates gracefully with ellipsis	Pass
5	Select Consultation Format	XPath = //android.widget.CheckBox[contains(@text, "Video Call")]	Video consultation chip toggles to active state	Video Call chip selected with active border styling	Pass
6	Choose Available Time Slot	XPath = //android.widget.TextView[@text="10:00 AM"]	Slot pill is selected and highlighted in primary theme color	Slot "10:00 AM" highlighted	Pass
7	Confirm Appointment Reservation	XPath = //android.widget.Button[@text="Confirm Booking"]	Modal closes, SnackBar confirms booking, appointments list updates	SnackBar displayed: "Appointment booked successfully"; list refreshed	Pass
Post-Condition: Appointment record created in PostgreSQL database appointments table with status Scheduled.					

Universal Wearables & QR Code Scanner Integration Test Case					
Test Case ID:		TC_FS_03		Test Designed By: Neeva Sunish Mathew / QA Lead	
Test Priority:		High		Test Designed Date: 07/10/26	
Module Name:		Wearable IoT & Smartwatch Sync		Test Executed By: Ms. Nimmy Francis / Automation Engineer	
Test Title:		Device QR Scan, BLE Discovery & Audio Filter		Test Execution Date: 07/10/26	
Description: Verify the Wearable Scanner screen, ensure paired audio accessories (e.g. Noise Buds) are excluded from smartwatch auto-connect to prevent GATT timeouts, and validate direct Bluetooth Low Energy discovery.					
Pre-Condition: Bluetooth adapter is enabled on Android device A059; phone has paired Bluetooth audio earbuds (Noise Buds VS402).					
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Open Smartwatch QR Scanner	XPath = //android.widget.Button[@content-desc="Scan Device QR"]	Scanner screen opens with live GPay-style viewfinder reticle	Screen opens titled "Scan Smartwatch QR" with active camera preview	Pass
2	Verify Audio Earbud Exclusion (Regression Check)	Paired Device: "Noise Buds VS402"	Audio peripheral is identified and skipped; no 15s GATT timeout	Earbuds bypassed by isAudioDevice() check; no connection freeze	Pass
3	Verify Direct Bluetooth Option Visibility	XPath = //android.widget.TextView[contains(@text, "Scan via Bluetooth")]	"📶 Or Scan via Bluetooth (No QR needed)" button is visible	Button rendered cleanly below the camera action button	Pass
4	Launch Real-time Bluetooth Radar	XPath = //android.widget.Button[contains(@text, "Scan via Bluetooth")]	Navigates to UniversalScanScreen and starts BLE scanning	BLE Radar scan initiated with active pulsing animation	Pass
5	Execute Quick Demo Pairing	XPath = //android.widget.TextView[contains(@text, "Pair boAt")]	Emulates watch QR payload boat://connect?mac=DC:1B:44:A2:89:12	Success dialog: "Smartwatch Connected! boAt Wave Beat"	Pass
6	Verify Live Telemetry Stream	XPath = //android.widget.TextView[contains(@text, "BPM")]	Real-time heart rate stream updates dynamically	Live Heart Rate tile displays stream (72–78 BPM)	Pass
Post-Condition: Smartwatch device adapter is registered to WearableManager and paired telemetry is synchronized with the backend.					

Digital Health Twin Vitals Logging & Analysis Test Case						
Test Case ID:		TC_FS_04		Test Designed By: Neeva Sunish Mathew / QA Lead		
Test Priority:		Medium		Test Designed Date: 07/10/26		
Module Name:		AI Health Twin & Daily Tracking		Test Executed By: Ms. Nimmy Francis / Automation Engineer		
Test Title:		Daily Metric Submission & Score Calculation		Test Execution Date: 07/10/26		
Description: Verify daily health metric entry (water intake, sleep duration, mood score) and confirm aggregate health score updates on the dashboard.						
Pre-Condition: Patient session is active; previous vitals history exists in health_tracker table.						
Step	Test Step	Test Data		Expected Result	Actual Result	Status
1	Open Daily Tracker Screen	Bottom Navigation / Tracker		Tracker form displays current daily logs	Log inputs for Water, Sleep, Mood, and Pain rating loaded	Pass
2	Increment Water Intake	XPath = //android.widget.Button[@content-desc="+250ml"]		Water count increments by 250ml	Daily water metric increased from 1500ml to 1750ml	Pass
3	Update Sleep Duration	XPath = //android.widget.EditText[contains(@hint, "Sleep")], Value: 7.5		Sleep field accepts numeric value 7.5	Field populated with 7.5 hours	Pass
4	Save Daily Log	XPath = //android.widget.Button[@text="Save Today's Log"]		API POST /api/health-tracker returns 200 OK	Backend returns success confirmation; metrics persisted in DB	Pass
5	Verify Digital Twin Score	XPath = //android.widget.TextView[contains(@text, "/100")]		Overall health score recalculates based on new entries	Health Score updated from 82 to 85	Pass
Post-Condition: Daily tracking entry saved to PostgreSQL and reflected on the Patient Digital Twin chart.						

Caregiver Dependent Multi-Profile & Medication Management Test Case					
Test Case ID:		TC_FS_05		Test Designed By: Neeva Sunish Mathew / QA Lead	
Test Priority:		High		Test Designed Date: 07/10/26	
Module Name:		Caregiver & Multi-Dependent Portal		Test Executed By: Ms. Nimmy Francis / Automation Engineer	
Test Title:		Dependent Profile Switch & Medication Scheduler		Test Execution Date: 07/10/26	
Description: Verify caregiver multi-dependent switching, dependent clinical vault access, and medication reminder dosage scheduling without interface regressions.					
Pre-Condition: Caregiver session authenticated (caregiver@femsphere.health), multiple dependents linked in PostgreSQL dependents table.					
Step	Test Step	Test Data	Expected Result	Actual Result	Status
1	Log in as Caregiver	XPath = //android.widget.TextView[@text="Marcus (Caregiver)"]	Redirects to Caregiver Portal (/caregiver-dashboard)	Caregiver dashboard active with dependent selector	Pass
2	Switch Active Dependent	Dropdown Selector, Value = "Maya (Daughter - 6y)"	Active view re-binds to selected dependent's health history	Dependent profile switched to Maya; records updated	Pass
3	Open Medication Reminders	XPath = //android.widget.TextView[@text="Medications"]	Displays scheduled prescriptions and dosages	Medication tracker screen loaded with daily dose list	Pass
4	Add New Medication Schedule	Medication: "Amoxicillin 250mg", Frequency: "Twice Daily"	Input modal validates fields and submits via POST /api/caregivers/medications	Medication added to schedule list; reminder timer active	Pass
5	Verify Dependent Medical Records Vault	XPath = //android.widget.TextView[@text="Medical Vault"]	Displays diagnostic lab reports and AI OCR extracted biomarker cards	Lab reports list rendered with download and OCR inspection actions	Pass
Post-Condition: Caregiver schedule synced; medication reminder saved and active in background alert manager.					

Execution Verdict: ALL TEST SUITES PASSED

5
Total Test Cases

5
Passed (100%)

0
Failed (0%)

3
Bugs Closed